

# HAOHAN WU

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## EDUCATION

**Stanford University**  
*B.S. Mathematics*

Stanford, CA  
*June 2027*

## RESEARCH EXPERIENCE

**Stealth AI Startup**  
*Researcher*

Los Altos, CA  
*Jun 2026 – Present*

- Designed and built an LLM-agent evaluation benchmark end to end as sole owner (~450 commits); sole-author manuscript in preparation.
- Proposed and prototyped a reflection component for the production backend; designed and ran the experiments.
- Ran controlled experiments on an in-house learning system; presented results at weekly company-wide demos.

## RELEVANT COURSEWORK

*Completed:* EE 364A Convex Optimization; EE 263 Matrix Methods: Singular Value Decomposition; MATH 171 Real Analysis; MATH 131P Partial Differential Equations; MATH 151 Introduction to Probability Theory; MATH 51–53 (multivariable calculus, linear algebra, ODEs)  
*Autumn 2026:* MATH 230A Theory of Probability I; CS 238 Decision Making under Uncertainty

## PROJECTS

**Optimal Ban Selection in *League of Legends* (in progress)**  
*Python (pandas, statsmodels), SQLite; Riot Match-V5 API*

Summer 2026 – Present

- Formalized ban value as a counterfactual: a ban substitutes an opponent rather than removing one. Built a resumable Match-V5 ingest, 182k ranked games over six tiers.
- Estimated the substitution effect from gold at 15:00 on champion, patch, side and autofill, clustered on match.
- Recovered the champion-by-champion interaction by method of moments on cell means net of sampling variance, with placebo and jackknife checks.
- Priced the proxy: gold at 15:00 and win rate rank-correlate 0.48, attenuation-corrected.

**One-Dimensional Heat Equation: Analysis and Implementation**  
*Class paper and code; Python (NumPy, Matplotlib)*

Mar 2025

- Derived the explicit FTCS scheme from second-order Taylor expansions; implemented a vectorized solver with configurable boundary and initial conditions.
- Validated against analytical solutions and a dynamic-boundary case.

**Logistic Regression from Scratch on Ancestry Data**  
*Python (NumPy, SciPy)*

Mar 2025

- Implemented logistic regression by maximizing log-likelihood with gradient ascent, no ML libraries.
- Ingested ~10k individuals' genetic records from CSV (type normalization, missing-value handling).

## OTHER EXPERIENCE

**Joyme**, AIGC Product Management Intern (remote)  
**Arbor Live**, Marketing and Social Media Coordinator, Stanford

Jul 2025 – Sept 2025  
Sept 2025 – Present

## SKILLS

*Scientific computing:* Python (numpy, pandas, cvxpy, PyTorch, scikit-learn), Julia; L<sup>A</sup>T<sub>E</sub>X; Git

## ADDITIONAL

*Languages:* English, Mandarin (fluent). *Interests:* music production and sound design (Ableton Live); 3D graphics (Blender).